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Title	Shell velocities in shell ellipticals									
WEAVE related surveys	None									
Summary	We propose to confirm shell-like kinematics in elliptical galaxies using the high velocity resolution of the WEAVE LIFU. In several prototype shell galaxies, WHT/ISIS long-slit spectroscopy revealed spatially-narrow velocity features. They were tentatively interpreted as kinematic shells, or nuclear kinematic signatures of the accretion events that create photometric shells visible in the outer parts. With the 3D capabilities of the WEAVE LIFU we will obtain full 2D kinematic maps and confirm or disprove kinematic shells. We also aim to exploit our preliminary results that WEAVE can obtain kinematic data on outer shells in the sample to distinguish between radial and tangential orbits, key inputs on the previous merger history. With the large wavelength coverage in LR mode, we will explore stellar population signatures in the nuclear shells such as local variations in age and alpha/Fe, and see if the distribution of emission lines also traces shell morphologies.									
Science and technical cases	1727460847_shells_25A_SciJus.pdf									
Focal-plane mode	WEAVE LIFU									
Spectrograph mode	Low-resolution blue+red,High-resolution green+red									
Spectral binning	x1									
Requested time	22.0 hours (Dark)									
Target	RA	Dec	Equinox	Band=Mag.	PROGTEMP	OBSTEMP	Setup & exposures	Observing conditions		
NGC0474NUC	01:20:06.70	+03:24:55.0	J2000	V=11.51	63331	IBDEC	LIFU HR_GR SpecBinBR_1 6x1020s	See.:≤1.5" Trans.:≥0.7 Elev.:≥38.7"(X=1.6) MoonDist.:≥30° SkyBright.:≥21.0Vmag/r ² (grey)		
NGC0474ESH	01:20:20.00	+03:25:08.1	J2000	SB_V=23.9	43331	IBDEB	LIFU LR_BR SpecBinBR_1 6x1020s	See.:≤1.5" Trans.:≥0.7 Elev.:≥38.7"(X=1.6) MoonDist.:≥30° SkyBright.:≥21.5Vmag/r ² (dark)		
NGC0474NSH	01:20:09.35	+03:26:06.7	J2000	SB_V=23.9	43331	IBDEB	LIFU LR_BR SpecBinBR_1 6x1020s	See.:≤1.5" Trans.:≥0.7 Elev.:≥38.7"(X=1.6) MoonDist.:≥30° SkyBright.:≥21.5Vmag/r ² (dark)		
NGC0474WSH	01:19:55.61	+03:23:13.9	J2000	SB_V=23.9	43331	IBDEB	LIFU LR_BR SpecBinBR_1 6x1020s	See.:≤1.5" Trans.:≥0.7 Elev.:≥38.7"(X=1.6) MoonDist.:≥30° SkyBright.:≥21.5Vmag/r ² (dark)		
NGC3656NUC	11:23:38.70	+53:50:31.9	J2000	V=12.4	63331	IBDEC	LIFU HR_GR SpecBinBR_1 6x1020s	See.:≤1.5" Trans.:≥0.7 Elev.:≥38.7"(X=1.6) MoonDist.:≥30° SkyBright.:≥21.0Vmag/r ² (grey)		
NGC3656SH	11:23:38.91	+53:50:18.1	J2000	V=12.4	43331	IBDEB	LIFU LR_BR SpecBinBR_1 6x1020s	See.:≤1.5" Trans.:≥0.7 Elev.:≥38.7"(X=1.6) MoonDist.:≥30° SkyBright.:≥21.5Vmag/r ² (dark)		
NGC5982	15:38:39.70	+59:21:20.0	J2000	V=11.3	43331	IBDEB	LIFU LR_BR SpecBinBR_1 6x1020s	See.:≤1.5" Trans.:≥0.7 Elev.:≥38.7"(X=1.6) MoonDist.:≥30° SkyBright.:≥21.5Vmag/r ² (dark)		
NGC5982	15:38:39.70	+59:21:20.0	J2000	V=11.3	63331	IBDEC	LIFU HR_GR SpecBinBR_1 6x1020s	See.:≤1.5" Trans.:≥0.7 Elev.:≥38.7"(X=1.6) MoonDist.:≥30° SkyBright.:≥21.0Vmag/r ² (grey)		
NGC7010NE	21:04:41.05	-12:19:54.1	J2000	V=14.5	63331	IBEEC	LIFU HR_GR SpecBinBR_1 6x1020s	See.:≤1.5" Trans.:≥0.7 Elev.:≥33.7"(X=1.8) MoonDist.:≥30° SkyBright.:≥21.0Vmag/r ² (grey)		
NGC7626	23:20:42.29	+08:13:02.5	J2000	V=11.11	43331	IBDEB	LIFU LR_BR SpecBinBR_1 6x1020s	See.:≤1.5" Trans.:≥0.7 Elev.:≥38.7"(X=1.6) MoonDist.:≥30° SkyBright.:≥21.5Vmag/r ² (dark)		
NGC7626	23:20:42.29	+08:13:02.5	J2000	V=11.11	63331	IBDEC	LIFU HR_GR SpecBinBR_1 6x1020s	See.:≤1.5" Trans.:≥0.7 Elev.:≥38.7"(X=1.6) MoonDist.:≥30° SkyBright.:≥21.0Vmag/r ² (grey)		

Scientific Justification

Unveiling the build-up of massive elliptical galaxies remains an important goal nearly 50 years after Alar Toomre's 1977 seminal paper on mergers. The problem has become increasingly interesting from work at low z (global scaling relations implying homogeneity; stellar populations showing old stellar ages) and at high redshift, eg., downsizing (Perez-Gonzalez+2008), up-sizing (Franx et al 2008); red-nuggets at high redshift (Damjanov+2011), and bulge-disk morphology at these redshifts (Costantin+2021). A favored emerging picture for massive ellipticals is the two-phase formation: an initial dissipational collapse possibly fed by cold flows (Dekel+2009) followed by an accretion phase $z \sim 1.5$ to 0, when the galaxy grows in size due to minor mergers. Size growth driven by minor mergers is supported by $z \sim 1$ velocity dispersions (Peralta de Arriba+2015) and merger-tree simulations (Frigo & Balcells 2017).

Shell galaxies are good laboratories to obtain clues about the nature of the second, accretion phase. Shells, or ripples, discovered by Malin & Carter (1983), were soon explained from radial minor mergers through the mechanism of phase wrapping in radial merger orbits (Quinn 1984). In this mechanism, shells mark the apocentres of accreted material as it oscillates through the centre. In the nineties, Carter and collaborators used the exquisite spectral resolution of the then new ISIS spectrograph on the WHT to study the core kinematics of several shell galaxies. They unveiled several narrow velocity features in the nuclei of two prototype shell galaxies: NGC 7626 (Balcells & Carter 1993) and NGC 474 (Hau+ 1996). See Figs 1, 3, 4.

Balcells & Carter (1993) hypothesized that these features might represent kinematic shells in the nuclear regions of galaxies with merger signatures such as shells and kinematically-decoupled cores (KDCs). If indeed linked to kinematic shells, these features would imply an undisrupted dynamical subsystem oscillating in the galaxy nucleus, quite unlike what's expected in phase-wrapped or spatially-wrapped shells, and, importantly, not immediately-expected in the two-phase model for the growth of massive ellipticals. In NGC 474 (see Alabi+ 2022), Balcells & Carter (unpublished) found strong suggestion of double-peaked l.o.s. velocity distributions which could however not be fully confirmed, despite a spectral resolution of $R=6000$ (see Fig. 4). Confirming double-peaked velocity distributions would open the door for a wide range of studies on the nature of the accretion process and the nature of the merger precursors, including stellar population diagnostics.

The radial/tangential nature of pre-merger orbits can be constrained by measuring line-of-sight velocities of outer shells. This is a difficult measurement due to the low surface brightness of shells. Balcells et. al (2000) published velocity measurements for the bright outer shell of NGC 3656, using WHT/INTEGRAL spectroscopy, but the uncertainty of the measurement prevented a diagnostic of the shell being radial (phase wrapping) or with significant tangential velocity (spatial wrapping). The other measurement is for the outer shell of NGC 474, using the small-format IFU of Keck/KCWI (Alabi+ 2020).

One WEAVE pointing at NGC 3656, obtained in director's discretionary time in 2024A2, which partially-covers the bright southern shell, has given evidence that this southern shell has a tangential velocity of ~ 80 km/s (Fig. 2, lower-right panel). Furthermore, these data suggest that WEAVE is capable of measuring l.o.s. velocities for other, fainter shells in this galaxy (Fig. 2).

The present proposal aims to exploit the NGC 3656 findings described above, to obtain 3D kinematic information for four prototype shell galaxies with high-quality HST imaging and ISIS multi-long-slit spectroscopy, both from previous collaborations (Carter and others). Spectroscopic information is key to the interpretation of the structure that we see at high spatial resolution in the ACS images, and will be used:

1. To obtain high-velocity resolution full 2D stellar kinematic maps of the nuclei of four galaxies being candidates to have kinematic shells. 2D maps of V , σ and high-order moments h_3 and h_4 will uniquely constrain subcomponents, including kinematic shells, providing a constrain

- on the dynamical age of the cores. Kinematics will be derived from the Mg and NIR CaII regions, which will provide information on dust effects on the kinematics (dust extinction is clearly present in some of the galaxy nuclei).
2. To obtain emission-line velocities for the gas being disrupted by the core of the remnant, and absorption-line velocities of the young stellar disks, to see whether the gas is being accreted in a disk, or as spherical accretion or inflow. The accretion of cold gas can lead to the direct formation of a young, kinematically decoupled component (e.g. NGC 5953: Falcon-Barroso+ 2006).
 3. To obtain deep R=2500 spectroscopy to infer l.o.s. velocities of brightest outer shells in the sample. This will provide dynamical confirmation of whether the shells are phase-wrapped or space-wrapped, and opening the way for detailed merger modeling.

For the nuclear kinematics, both R=10,000 (HR) and R=2500 (LR) data will be obtained. HR (Mg, and H α regions) will provide the resolving power required to map the kinematic substructures and any peculiar line-of-sight velocity distributions such as double-peaked distributions in NGC 474. LR will provide dynamical and stellar population information extracted from a wide wavelength range. Young population and recent starbursts (eg. NGC 3656) will be revealed, identifying the radial distribution of young populations (cf. the Cartwheel galaxy, Marcum et al 1992).

The proposed observations will advance our knowledge on the nature of the merger orbits and the contribution of the accreted material to build up the high-density nuclei and the outer envelopes of ellipticals. Also, we will advance on the knowledge of young stellar populations originating in merger events that are traced by shells.

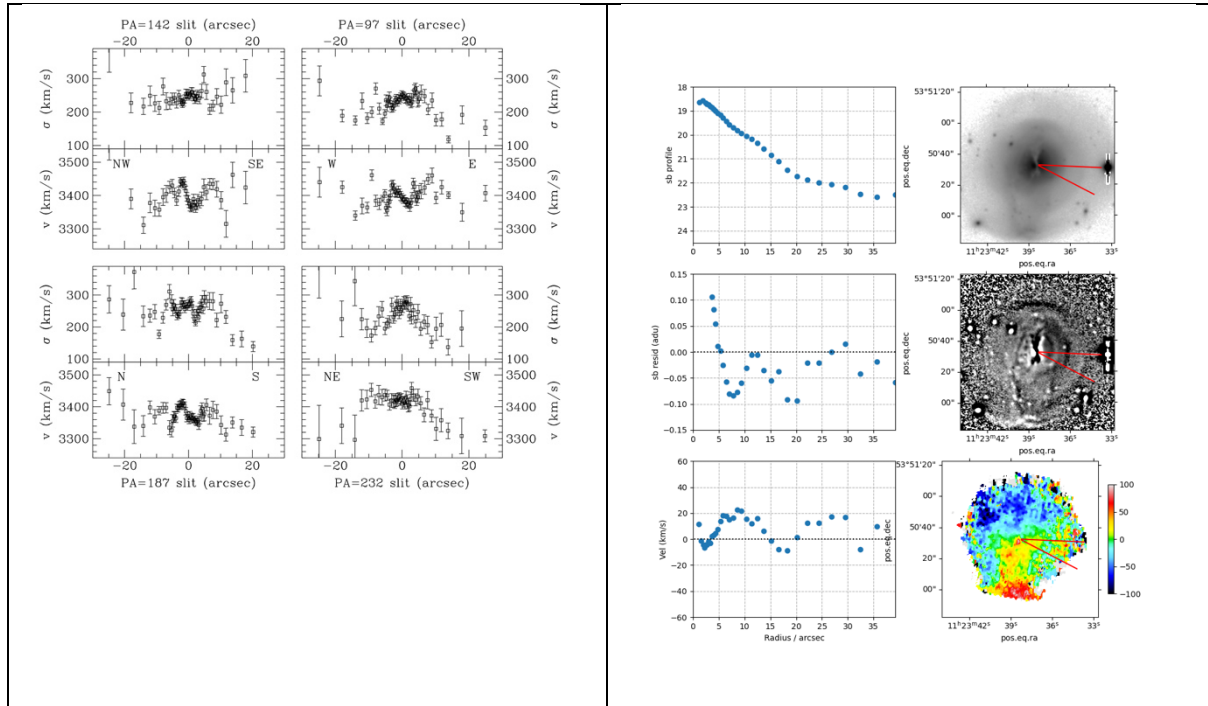


Figure 1 (left): Four velocity and dispersion profiles for NGC 7626, from Balcells & Carter (1993). Sharp velocity features ('kinematic shells') are found along position angles 97, 142, 187.

Figure 2 (right): velocity map (bottom-right) of NGC 3656, from discretionary WEAVE observations. The bright southern shell shows $v=+80$ km/s over systemic. And the fainter shell to the west (along the red wedge aperture) shows tantalizing velocity oscillations coinciding with the shell positions. Shell positions are seen on the top-right image, and better traced in the unsharp-masked image (middle-right). The left panels, top to bottom, show profiles along the red wedge aperture, of R-surface brightness; unsharp-masked brightness, and l.o.s. velocity, respectively.

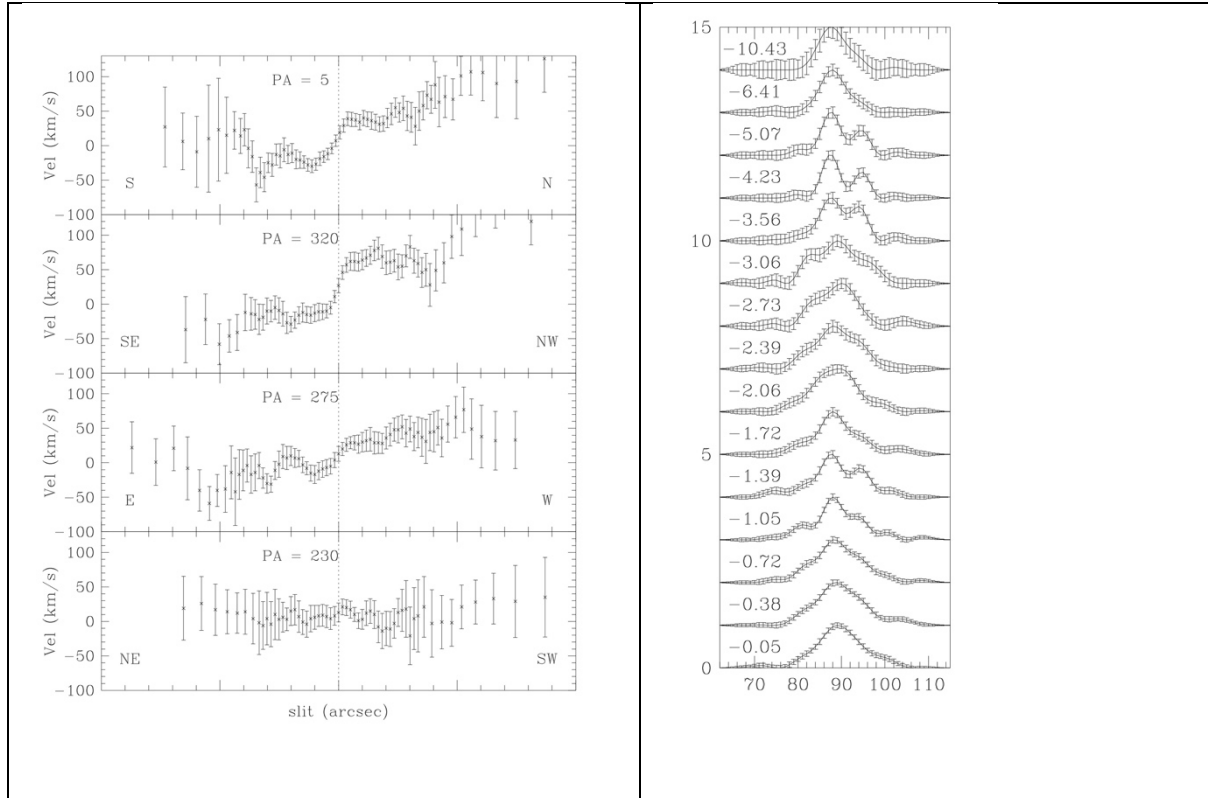


Figure 3 (Left): Four velocity profiles for NGC 474, from Hau et al (1996). Sharp velocity features are found along position angles 5 and 275.

Figure 4 (Right): Line-of-sight velocity distributions (fv) for NGC 474. They correspond to positions from the center to $-10.43''$ along position angle 275. A strong suggestion of double-peaked fv appears on 3 bins from $3.5''$ to $5.07''$. The fv were determined using unresolved gaussian decomposition (Kuijken & Merrifield 1993). The x-axis gives line-of-sight velocities in units of 25.81 km/s . The higher WEAVE/LIFU HR resolving power, as compared with the ISIS data used in this figure, will clearly help in resolving any separate components.

Technical Case

Sample

The sample comprises four ellipticals/S0 well studied by our group. In particular, we have previous HST/ACS imaging and WHT/ISIS spectroscopy. The four galaxies have shells and nuclear peculiar kinematics (Balcells & Stanford 1988; Balcells & Carter 1993; Oosterloo et al. 1994; Hau et al. 1996; Sikkema et al. 2006, 2007) hence they are the best candidates for this first study with WHT/WEAVE LIFU.

Observational strategy

The WEAVE LIFU is the instrument of choice for this project owing to its fantastic FOV, its high throughput and its unique HR spectral resolution $R=10,000$. Its main draw-back is its coarse spatial sampling (2.6 arcsec fibres), which will compromise the detection of narrow velocity features (typically 2 arcsec). VLT/MUSE provides an excellent spatial sampling but is not a good alternative owing to its low $R=2000$ @ 0.46 micron . Our strategy is to maximize the spatial sampling by taking dither-6 observations: the depth requirements for outer shells need 2-hr exposures in any case.

For the HR observations, the Green VPH, covering H-Beta, [OIII], Mg and Fe lines is an ideal region for both stellar and ionized-gas kinematics. The red VPH, centered on H-Alpha, will deliver exquisite emission-line profile information as well as sensitive star-formation indices and extinction estimates from H-Alpha / H-Beta ratios.

Exposure times are established by the SNR requirements inferred from our previous spectroscopy. We provide sample calculations for NGC 474, which has V-surface brightness of 19.5 mag/sq.as (radius of 8 arcsec, see Sikkema+2007). Our N474 longest exposure was $t_{\text{exp}}=9000$ s and gave $S/N=30$ per (spatial x spectral) pixel = (0.3sq.as x 26 km/s).

With WEAVE HR2 (green [red]), we have 0.090 [0.11] A/pix = 5.4 [6.6] km/s/pix, which oversamples (x6) the resolving element of 30/km/s but is overall beneficial to improve sky subtraction in the NIR Call triplet region. Given ~ 10 spectral pixels per Angstrom, $SNR=30$ per spectral pixel corresponds to $SNR=30 \times \sqrt{10} = 95$ per Angstrom. According to the WEAVE ETC (<https://astro.ing.iac.es/signalWEAVE/>), 7200 s are needed to deliver $SNR=95$ per Angstrom for $V=19.5$ mag/sq.as. We therefore need a standard dither-6 for HR observations.

For LR, exposure times are derived from the requirement to obtain velocities for outer shells. For Blue [Red] we have 0.30 [0.48] A/pix = 18 [20] km/s/pix, which again oversamples the resolving element of 120 km/s, this allows for post-sky-subtraction spectral binning. For NGC 3656 the outer shell has surface brightness of 22 R-mag/sq.as (Balcells+2001), whereas the outer shell of NGC 474 has 23.9 V-mag/sq.as (both Vega). For the fainter NGC474, using signalWEAVE, 2-h integration coadding 9 LIFU fibres yields S/N per A = 12.7, and a radial velocity error of 15 km/s, adequate to constrain radial vs tangential shell motion.

Dark time is necessary given the low surface brightness of the outer shells, whereas good seeing is needed for the tracing of the inner kinematic shells. Seeing requirements for each OB will be specified in Phase 2.

Processing and analysis

Basic processing will use the standard CPS and APS pipelines, but a number of additional tools will need to be used. Most important will be a cube reconstruction that maximizes spatial resolution, as described in Sanchez et al (2023), taking advantage of high-quality, HST as well as ground-based imaging of these well-studied galaxies.

Double-peaked velocity distributions will be investigated with non-parametric extraction methods such as BAYES-LOSVD (Falcon-Barroso & Martin 2021).

References:

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